A CUSTOMIZED CALYPSOX GSM PROJECT

Classic mobile hardware. A deeper view of GSM.

MiniOcom CalypsoX Lite is an experimental firmware customization. It brings classic handset functionality together with GSM network monitoring, anti-interception indicators, subscriber-identity test features, and SMS-format experiments.

Discover the features → Explore SMS formats

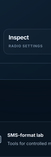
Experimental firmware · Engineering-focused tools · Classic GSM platform

GSM MONITOR / OVERVIEW ★ SAMPLE VIEW

RADIO ENVIRONMENT

Network insight

Serving cell and neighbor information



MOTOROLA C123

Monitor — CELL INFORMATION
Inspect — RADIO SETTINGS
Experiment — SMS FORMATS

ILLUSTRATIVE DESIGN — CalypsoUSSM

Network visibility — Serving and neighboring cell details

Experimental indicators — Security-related monitoring options

SMS-format lab — Tools for controlled message-format tests

WHAT IT CAN DO

Useful tools, made easier to understand.

The firmware adds a dedicated GSM Tools area and experimental options that help a developer or radio enthusiast observe how a compatible handset behaves on a GSM network.

GSM Monitor

View information reported by the serving cell, the network, and neighboring cells in one engineering screen. It is intended to make radio conditions easier to inspect.

NETWORK INFORMATION

Cell Lock

Access the cell-lock controls for experiments with a selected GSM channel or cell. Locking can affect registration and service, and results depend on radio conditions and network behavior.

RADIO EXPERIMENTS

Cipher Mode Tools

Inspect the firmware's cipher-mode options and related network behavior. These controls do not override the operator's network policies or guarantee that a particular encryption mode will be used.

GSM ENGINEERING

Anti-Interception Indicators

Optional phone-side monitors watch for silent-SMS indications, repeated registration or location-update patterns, suspicious serving-cell changes, identity requests, ciphering events and silent-call paging.

EXPERIMENTAL MONITORING

SMS Format Testing

Choose from standard SMS and several specialized formats to study message handling and compatibility on controlled test devices and SIM cards.

MESSAGE TESTING

Refined Handset Interface

Custom menu organization and updated menu and submenu icons give the added engineering tools a clear place in the classic handset interface.

INTERFACE CUSTOMIZATION

COMPLETE GSM TOOLS DIRECTORY

The full menu, explained in plain language.

GSM Tools is a dedicated application in the phone's main menu. It brings together GSM network information, radio-security indicators, subscriber-identity test features, and selected phone-management functions.

GSM Monitor

GSM Monitor is a live overview of the phone's current GSM serving cell and nearby cells. The information is refreshed while the monitor is open.

SERVING-CELL AND NETWORK INFORMATION INCLUDES

ARFCN	The GSM radio channel number used by the serving cell.
Band	The GSM band category associated with that channel, such as GSM 900, E-GSM, DCS, or PCS.
MCC and MNC	The country and mobile-network identifiers broadcast by the network.
LAC	The Location Area Code identifying the current location area.
CID	The Cell ID identifying the serving cell within the network.
BSIC	The Base Station Identity Code used to distinguish GSM base stations.
TA	Timing Advance, a radio timing value associated with the phone's link to the serving cell.
RXLEV	The reported received signal level.
RXQ	The reported radio signal quality.
C1 and C2	GSM cell-selection and reselection criteria values.
RLT	The Radio Link Timeout value reported for the serving connection.
GSM/GPRS	Indicates the reported serving-cell radio type.
Lock status	Shows whether an ARFCN lock is active or switched off.

NEIGHBOUR-CELL INFORMATION

Neighbour-cell information shows nearby cells reported by the network. For each displayed neighbour, the monitor can show its ARFCN, received signal level, BSIC, C1 and C2 values. The current screen is designed to show up to six neighbour-cell entries.

GSM Monitor gives a quick view of the serving-network, current radio conditions and neighbouring GSM cells visible to the phone.

C2 Monitor

C2 Monitor is a security-event counter screen. It displays totals recorded by the firmware for categories of suspicious or unusual GSM activity.

EVENT COUNTERS

Silent SMS	Events identified as silent-SMS activity.
Silent Call	Events associated with suspected silent-call activity.
Paging Attacks	Unusual paging events counted by the monitor.
IMSI Catcher	Events associated with identity requests that may indicate an attempt to identify or track the subscriber.
GSM Interceptor	Suspicious serving-cell changes or related radio events.
Location Tracking	Repeated registration or location-update patterns that may indicate tracking behavior.

The screen also indicates whether the interception monitor is active. The counters are intended as an overview of events observed by the phone, not as proof that an attack or interception is present. The detection methods use limited data from the normal network behavior may sometimes look suspicious and some hostile activity may not be detected.

Anti Intercept

Anti Intercept is an optional set of phone-side security monitors. It has an Enable All control and independent controls for the individual detectors.

AVAILABLE MONITORING CATEGORIES

Silent SMS Detection	Identifies incoming silent-SMS indications and can display a short alert.
UnPag / Location Tracking	Looks for repeated registration or location-update patterns that may be associated with attempts to track the phone.
GSM Interceptor / Proximity Alert	Warns about suspicious serving-cell changes, identity requests, and combinations of call and ciphering events.
Silent Call Paging	Identifies paging patterns that may be associated with silent calls and can display an alert.
Alert Buzzer	Enables or disables the audible warning buzzer for alerts.

Enable All turns the individual detection categories on together. The individual categories can also be controlled separately.

Example alert labels: SILENT SMS, SILENT CALL, CELL CHANGE, CELL RISK, IMSI REQ, or UNPAG TRACK. These labels are warnings about observed network behavior. They do not establish the identity of a transmitter, prove that an IMSI catcher is present, or guarantee detection of every tracking or interception attempt.

SMS Control

SMS Control is a configurable remote callback/notification feature.

MAIN SETTINGS

Secret Message	The private text that must be present in the incoming SMS.
Allowed Sender Number	The sender number that is permitted to issue the request.
Callback Number	The phone number that the handset calls when a valid request is received.

When enabled, the feature accepts a request only when the incoming message matches the configured secret and comes from the configured sender. A matching request causes the handset to place a voice call to the configured callback number.

Silent Mod is an optional display/notification behavior for this feature. When enabled, a matching control SMS is handled without the usual new-SMS notification. It is intended for the configured control message, not as a general setting that hides every incoming SMS.

Clear Call Logs

Clear Call Logs manages the phone's recent call-history lists.

CALL-HISTORY LISTS

Last dialled calls · Last received calls · Last missed calls

The menu includes immediate clearing, automatic clearing after calls, and a daily time-based clearing option.

This feature concerns call-history records; it is not a general deletion tool for SMS messages or phonebook contacts.

Cell Lock

Cell Lock controls whether the phone follows normal automatic GSM channel selection or is restricted to a selected ARFCN.

AVAILABLE OPTIONS

Manual	Selects a specific ARFCN as the channel to which the phone should remain locked.
Auto	Releases the manual channel restriction and returns to automatic network/channel selection.

The GSM Monitor displays the current lock state and the locked ARFCN when one is active. Cell Lock is a radio-channel restriction; it is not a guarantee that a particular cell will remain available or provide service.

CIPHER Mod

Displays GSM ciphering status and the reported A5 algorithm when that information is available.

CIPHER-STATUS INDICATION

Closed-lock symbol	Indicates that GSM ciphering is active.
Open-lock symbol	Indicates no GSM ciphering or a weak-cipher warning.
Algorithm number	The number beside the symbol identifies the reported A5 algorithm when available.
A5/0	Means that GSM traffic is not ciphered.
A5/2	Is treated as a weak-cipher warning.
A5/3 to A5/1	In Auto mode, a change from A5/3 to A5/1 is treated as a possible cipher downgrade and can produce a warning.

The warning can appear as a short on-screen cipher alert.

A5/3 status: Beta — not working yet.

IMSI Changer

IMSI Changer provides controls for an alternative subscriber identity on a physical SIM profile.

AVAILABLE FUNCTIONS

Set IMSI	Stores a user-provided IMSI value.
Random IMSI	Creates an alternative IMSI value while retaining the first five digits of the current SIM IMSI as the network prefix.
Enable	Activates the stored IMSI override.
Disable	Returns identity use to the physical SIM's normal IMSI.

The IMSI Change function allows the user to change the IMSI value SIM registration number stored on the SIM card), while keeping the same prime number (MSISDN).

Detach IMSI/TMSI

Covers GSM mobility-management detach and related subscriber-registration and radio-access concepts.

GSM CONCEPTS

IMSI Detach	A GSM mobility-management procedure that allows a mobile device to inform the cellular network that it is no longer available. It is normally performed when a phone is switched off or disconnected from the network. The network can then update the subscriber's registration status.
IMSI Attach Flood	Refers to excessive IMSI Attach or location-update signaling requests. IMSI Attach is normally used when a mobile device becomes available and registers with the network. Repeated signaling attempts can increase the workload on network registration procedures and consume signaling resources.
RACH Flood	Refers to excessive Random Access Channel (RACH) requests. RACH is the GSM uplink channel used by mobile devices to request initial access to network resources. Excessive requests can increase channel contention, cause collisions, and delay legitimate devices attempting to access the network.
Paging-race condition	A timing condition in the GSM paging procedure that may affect the delivery of incoming phone calls or SMS.

MAIN DIFFERENCES

IMSI Detach	Changes the subscriber's availability status.
IMSI Attach Flood	Involves repeated subscriber-registration signaling.
RACH Flood	Involves excessive radio-access requests.

SIM Clone OTA / Virtual SIM

Despite the menu name, the supplied description identifies this feature as a local Virtual SIM/host-card profile editor.

PROFILE FIELDS

IMSI	Subscriber identity.
TMSI	Temporary subscriber identity.
Ki	GSM authentication/ciphering key value.
Sequence number	The cipher key sequence value.
ARFCN	Radio channel selection value.
MCC and MNC	Mobile-network identifiers.
LAC	Location-area information.
SMSC	SMS service-centre number.

The profile can be stored and activated as a virtual subscriber/host-card identity. The menu also provides a disable function that returns the handset to use of the physical SIM.

The profile is intended for controlled GSM laboratory and compatibility testing; valid subscriber credentials and network-side authorization are still required for successful service.

Dynamic IMEI

Dynamic IMEI is an enable/disable setting for automatic IMEI changing.

HOW IT IS DESCRIBED IN THE SUPPLIED TEXT

When active, the firmware generates a fresh IMEI value for a new mobile-originated voice call or SMS transaction.

The option is intended for controlled device-identity testing. It changes the handset identity presented for those transactions and should not be mistaken for ordinary privacy encryption.

Random IMEI

Random IMEI generates a new 15-digit IMEI value.

The menu provides an option to apply the generated value to the handset. Applying it stores the value; the supplied text states that a phone restart is not required.

SMS FORMAT LAB

More than a standard text message.

The SMS composer has an SMS Type selector. It lets the user choose the format or behavior of the message being composed. The available choices in this source version are described below.

Controlled testing only. Use your own test SIM and devices, or obtain explicit authorization. The SMS Crash item describes a fault affecting vulnerable Android devices; do not send malformed-text messages to other people or live services.

NORMAL SMS

Standard text message

Normal SMS

A standard text message sent through the phone's ordinary SMS function. This is the normal default type for a newly composed message.

CLASS 0 SMS

Flash SMS display behavior

Class 0 SMS · Flash SMS

A Class 0 message, commonly known as a Flash SMS. On compatible recipient phones it is presented immediately on the display instead of behaving like a regular message that is first placed in the normal inbox. The exact display and storage behavior depends on the recipient handset and network.

WAP PUSH URL

Service indication (SI)

WAP Push URL · Service Indication (SI)

Sends a WAP Push Service Indication containing a URL. Compatible handsets can present the received push as a link or notification that the recipient can open with a WAP/browser client.

WAP PUSH SL

Service Loading

WAP Push Service Loading (SL)

Sends a WAP Push Service Loading message containing a URL. On compatible handsets, this asks the WAP client to load the specified content rather than presenting only a normal text message. Whether it is acted upon depends on the recipient's WAP client and security settings.

S@T SETUP-CALL & USSD EXECUTE

SIM Application Toolkit message format

S@T Setup-Call · USSD Execute

Sends a SIM Application Toolkit (SIAT) call-setup message. On a compatible SIM and handset, it requests a call-setup action using the number supplied for the message. The recipient device and SIM determine whether the request is supported or accepted.

SMS TYPE 0

Non-display / quiet SMS category

SMS Type 0

A Type 0 SMS is a non-display/quiet SMS category rather than an ordinary visible text message. In this firmware, the editor text is not used for this type and a delivery-status report is requested. Recipient and network behavior can vary.

SMS CRASH

Malformed WAP Push security-impact entry

SMS Crash · WAP Push · Android telephony

The supplied text states that a single malformed WAP Push SMS can crash com.android.phone on Android, dropping active calls (including emergency calls), with zero interaction from the victim.

ABOUT THE PROJECT

A customized firmware for hands-on GSM learning.

MiniOcom Calypsox Lite brings together a classic Calypso handset platform and a collection of firmware-level experiments. It is aimed at developers, radio enthusiasts, and researchers who want to explore legacy mobile technology in a lab setting.

Project type	Community customization
Base source tree	FreeCalypso / Tourmaline
Primary focus	GSM tools and SMS testing
Release status	Experimental